

Siyu Chen

Email: scanlonchan@gmail.com; Tel: (+86) 17600992592

Education

Beijing Forestry University

Beijing, China

B.E. in Electronic Information Science and Technology

09.2020-07.2024

Work Experience

Tsinghua University

Department: Industrial Engineering

Beijing, China

Research Assistant

Mentor: Xiwen Bai & Yilin Wang

01.2025-03.2026

1. A Study on Transfer Learning Based on Compression Fractures.
2. A Two-Stage Model Cascade Study Based on Compression Fractures.
3. Research on Multi-Task Learning Based on Compression Fractures.
4. Application of YOLOv13 in Port Infrastructure Feature Extraction.
5. System Module Development for the Intelligent Cargo Packaging System.

Publication

No.	Journal Articles
J6	An interpretable machine learning framework for preoperative deep vein thrombosis risk stratification based on thrombus size in fracture patients <i>(Under Review)</i> Yilin Wang*, Siyu Chen*, et al. <i>Scientific Reports</i> 2026
J5	VerteX-ID: A Multi-Task Learning Framework for Osteoporotic Compression Fracture Detection <i>(Under Review)</i> Yilin Wang*, Siyu Chen*, Hongwei Zhuang*, et al. <i>IJSE Transactions on Healthcare Systems Engineering</i> 2026
J4	A Two-Stage Approach for the Identification and Classification of Osteoporotic Vertebral Compression Fractures [paper] Yilin Wang*, Siyu Chen*, Hongwei Zhuang*, et al. <i>Medical Physics</i> 2026
J3	Identifying acute thoracolumbar vertebral compression fractures from low-quality small-sample X-ray images: a transfer learning-based approach [paper] Yilin Wang*, Weijun Li*, Siyu Chen*, et al. <i>IEEE Journal of Biomedical and Health Informatics</i> (Q1, 'IF=7.7) 2025
J2	Waveform purification of ultrasonic signals from xylem cavitation in tree species based on U-MFCC feature extraction and voiceprint recognition models [paper] Siyu Chen, et al. <i>Computers and Electronics in Agriculture</i> 2025
J1	Soft X-ray image recognition and classification of maize seed cracks based on image enhancement and optimized YOLOv8 model [paper] Siyu Chen, et al. <i>Computers and Electronics in Agriculture</i> (Q1, 'IF=10.3) 2024

* Co-first authors with equal contributions

No.	Invention Patents
-----	-------------------

P4	A Method, Apparatus, Device, and Computer Storage Medium for Packaging. CN121317420A Siyu Ding, ..., Siyu Chen . 2025
P3	Ultrasonic Cavitation Signal Purification for Xylem and MFCC Feature Enhancement Extraction Method. CN120234602A Xiangxue Zhang; Yifeng Li; Siyu Chen ; et al. 2025
P2	Soft X-ray Imaging of Internal Cracks in Corn Seeds: YOLOv8 Neural Network Recognition Method. CN117218544A Xiangxue Zhang; Siyu Chen ; et al. 2023
P1	Automatic Detection Device for Internal Cracks in Corn Seeds Based on Soft X-ray Image Recognition. CN117299597A Xiangxue Zhang; Siyu Chen ; et al. 2023
No.	Computer Software Copyright
C1	Plant Information Retrieval and Export Platform System. 2023SR0519773 Siyu Chen , 2022

Research Experience

An interpretable machine learning framework for preoperative deep vein thrombosis risk stratification	
<i>Project Leader</i>	1/2026-4/2026
➤ Developing a three-level DVT risk stratification framework based on thrombus size, integrating 62 clinical variables from 1,216 fracture patients. ➤ Combining LASSO feature selection with optimized CatBoost + SMOTE, achieving 96.67% accuracy, 96.67% F1-score, and 98.94% AUC-ROC using the first 41 selected features. ➤ Outperforming traditional risk scores and baseline machine learning models, with SHAP analysis revealing distinct pathological patterns across thrombus-risk levels.	
A Two-Stage Approach for the Identification and Classification of Osteoporotic Vertebral Compression Fractures [paper]	
<i>Project Leader</i>	07/2025-10/2025
➤ Proposing a two-stage cascade model for vertebral segmentation and fracture classification in X-ray images. ➤ Enhancing DeepLabv3+ for segmentation, improving Dice by 0.24% and reducing Hausdorff distance by 12.3%. ➤ Supporting both binary and three-class fracture classification, achieving average precision of 72.90 and 50.16, respectively. ➤ Outperforming single-stage models (53.51/39.59) and orthopedic surgeons (46.88/11.64) in binary/three-class classification.	
Identifying acute thoracolumbar vertebral compression fractures from low-quality small-sample X-ray images: a transfer learning-based approach [paper]	
<i>Project Leader</i>	03/2025-08/2025
➤ Developing a transfer learning model for acute thoracolumbar vertebral compression fracture detection from small, low-quality X-ray datasets. ➤ Integrating multi-scale edge and texture features with ViTDet and Faster R-CNN. ➤ Improving classification accuracy from 27% (manual) to 33% (model-based). ➤ Requiring 307.28 GFLOPs and 105.95M parameters, with 18.44 s inference time for the complete test set.	
Waveform purification of ultrasonic signals from xylem cavitation in tree species based on U-MFCC feature extraction and voiceprint recognition models [paper]	

- Proposing U-MFCC feature extraction, extending the MFCC frequency range to 0–10 kHz.
- Improving recognition accuracy by approximately 10% across metrics compared with original MFCC.
- Achieving an average 15% improvement in cavitation signal purification over commonly used clustering algorithms.

Soft X-ray image recognition and classification of maize seed cracks based on image enhancement and optimized YOLOv8 model [\[paper\]](#)

Project Leader

08/2023-12/2023

- Proposing a non-invasive imaging technique combining wavelet denoising, image standardization, bilateral filtering, and Laplacian sharpening.
- Optimizing YOLOv8, improving AP by 3.1%, recall by 4.3%, and mAP by 2.7%, while reducing computational cost by 1.7 GFLOPs.
- Achieving 99.66% accuracy, 99.87% precision, 99.48% recall, and 7.49 ms/frame detection time over 10 trials.

Academic service

Peer reviewer experiences:

- *IJSE Transactions on Healthcare Systems Engineering*, 09/2025;
- *Journal of Agricultural Science and Technology*, 05/2025;
- *Molecular & Cellular Biomechanics*, 01/2025;
- *Artificial Intelligence in Agriculture*, 08/2024;
- *Mathematical Biosciences and Engineering*, 01/2024.

Industry Experience

Intern | Easthome (Beijing) Consulting & Service Co., Ltd.

05/2022-06/2022

- As a project manager of a group, leading the members to develop a JAVA Android application.

Skills

- **Language:** Mandarin (native), English (fluent; TOEFL 5.0/6.0: R5.5/L5/S5/W5)
- **Compiler software:** Proteus, Electronic CAD, VS code, PyCharm;
- **Programming Languages:** C, C++, Java and Python;
- **Design/Development:** Analog integrated circuit design, Digital integrated circuit design, Single-chip microcomputer development; Application development.